

CreditWise Project Description

Project Title: An Explainable AI Framework for Loan Risk Assessment and Decision Support Using Machine Learning

Project Overview

This project aims to build an intelligent loan decision support framework using Machine Learning and Explainable Artificial Intelligence (XAI). The system predicts whether a loan application is likely to be approved while providing transparent explanations for every prediction using SHAP and LIME. It also estimates applicant risk and offers recommendations that can help improve loan eligibility. The objective is to support bank officials with accurate, explainable, and trustworthy AI-assisted decisions rather than replacing human decision makers.

Problem Statement

Many loan approval systems either depend on manual evaluation or use black-box machine learning models whose decisions cannot be easily interpreted. Our project addresses this challenge by combining predictive models with explainability techniques so that both applicants and financial institutions can understand the reasons behind approval or rejection.

Objectives

1. Develop accurate machine learning models for loan approval prediction.
2. Compare SHAP and LIME for explainable AI.
3. Perform applicant risk assessment.
4. Generate recommendation-based decision support.
5. Deploy the complete framework using Streamlit.

Technology Stack

Python, Kaggle, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, XGBoost, SHAP, LIME, Joblib, Streamlit, Git and GitHub.

Modules

- Dataset Collection
- Data Preprocessing
- Exploratory Data Analysis
- Feature Engineering
- Machine Learning Model Development
- Model Evaluation
- SHAP Explainability
- LIME Explainability
- Risk Assessment
- Recommendation Engine
- Streamlit Dashboard
- Research Paper and Documentation

Methodology

Dataset Collection → Data Preprocessing → Exploratory Data Analysis → Feature Engineering → Machine Learning Model Training → Model Evaluation → Best Model Selection → SHAP & LIME

Explainability → Risk Assessment → Recommendation Engine → Streamlit Dashboard.

Expected Deliverables

- Trained ML model
- Explainable AI framework
- SHAP and LIME comparison
- Risk assessment module
- Streamlit dashboard
- IEEE-style research paper
- Project report
- PowerPoint presentation
- GitHub repository

Novelty

Unlike conventional student projects that only predict loan approval, this framework compares two explainability techniques (SHAP and LIME), integrates risk assessment, generates recommendations, and provides an interactive decision support dashboard.

Area	Details
Domain	Artificial Intelligence, Machine Learning, Explainable AI, FinTech
Duration	8 Weeks
Development	Kaggle (ML) + VS Code (Streamlit)
Target Users	Banks and Financial Institutions
Output	Transparent loan approval decision with explanations and recommendations